

Finalizing GEMini update that:

1. Automates pedestal finding/setting
2. Filters noisy events from pedestal finding

These changes:

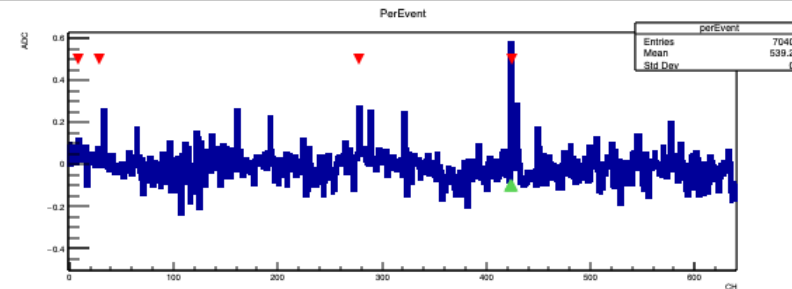
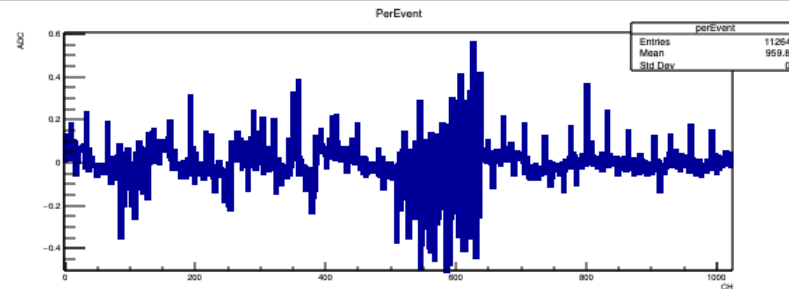
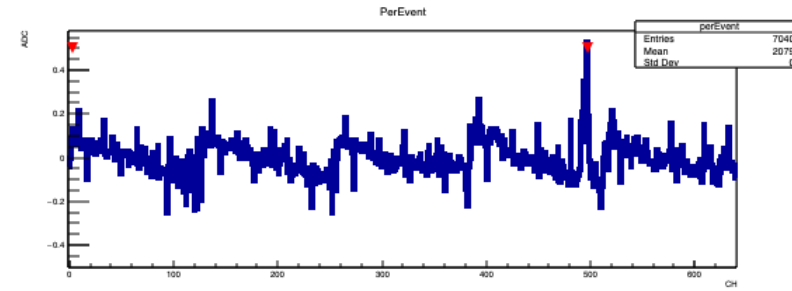
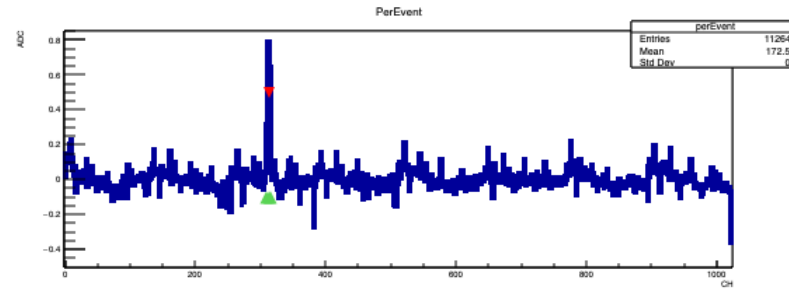
1. Make APV ringing nearly negligible
2. Centers data on median, including square waves.
3. Remove peaks that stay consistently above/below zero
4. Allow clear identification of peaks with amplitudes down to 300 ADC values

Method:

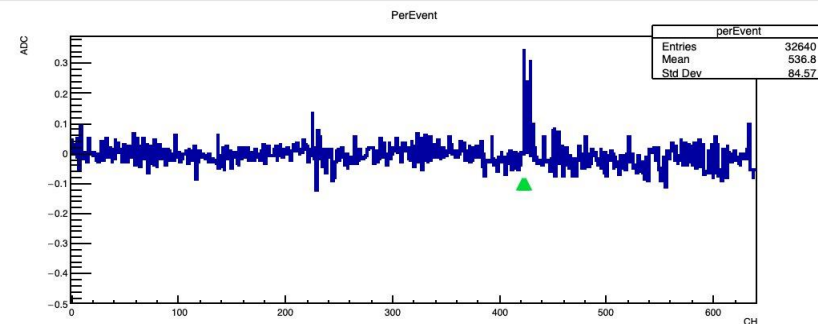
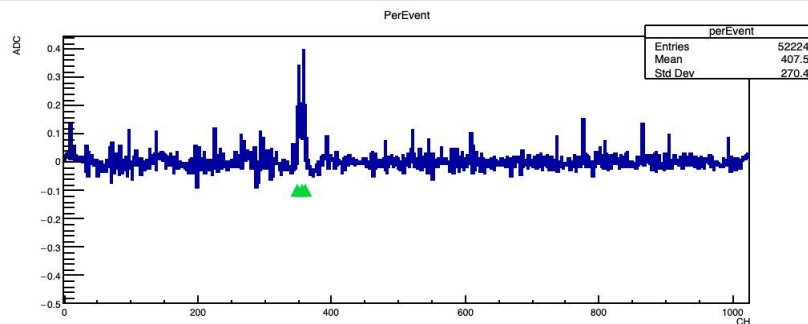
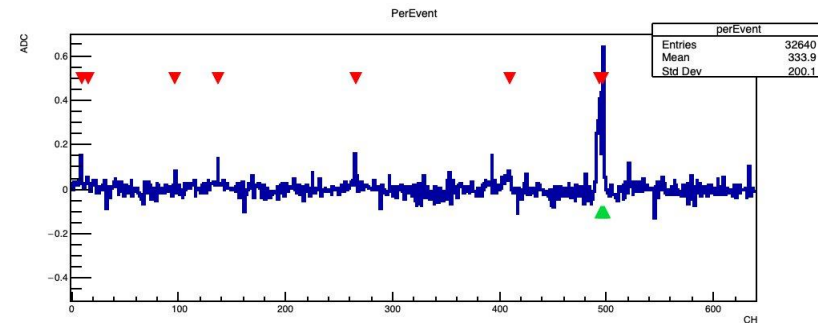
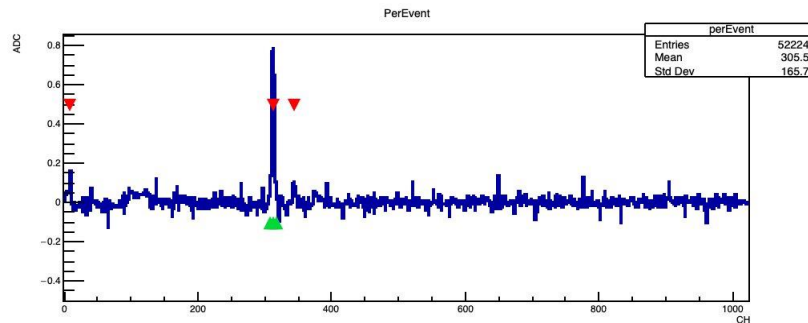
1. For each event
 - a) Finds the slope of ADC values between time samples (each channel on all 4 GEMs)
 - b) Averages all of the slopes (all channels and time-samples)
2. Identifies all events with slope < 300 ADC values / time sample (~ half of all events)
3. Fills a histogram with ADC values from the first time-sample of identified events.
4. Fits a gaussian curve with fixed range (1000-1800) compared to previous method (max value ± 30)
 - a) (filtering by slope removes 2nd unwanted peak by 800 ADC)
5. Automatically sets pedestals for that event.

Run 4582, Event 61, Left Side

Old Method

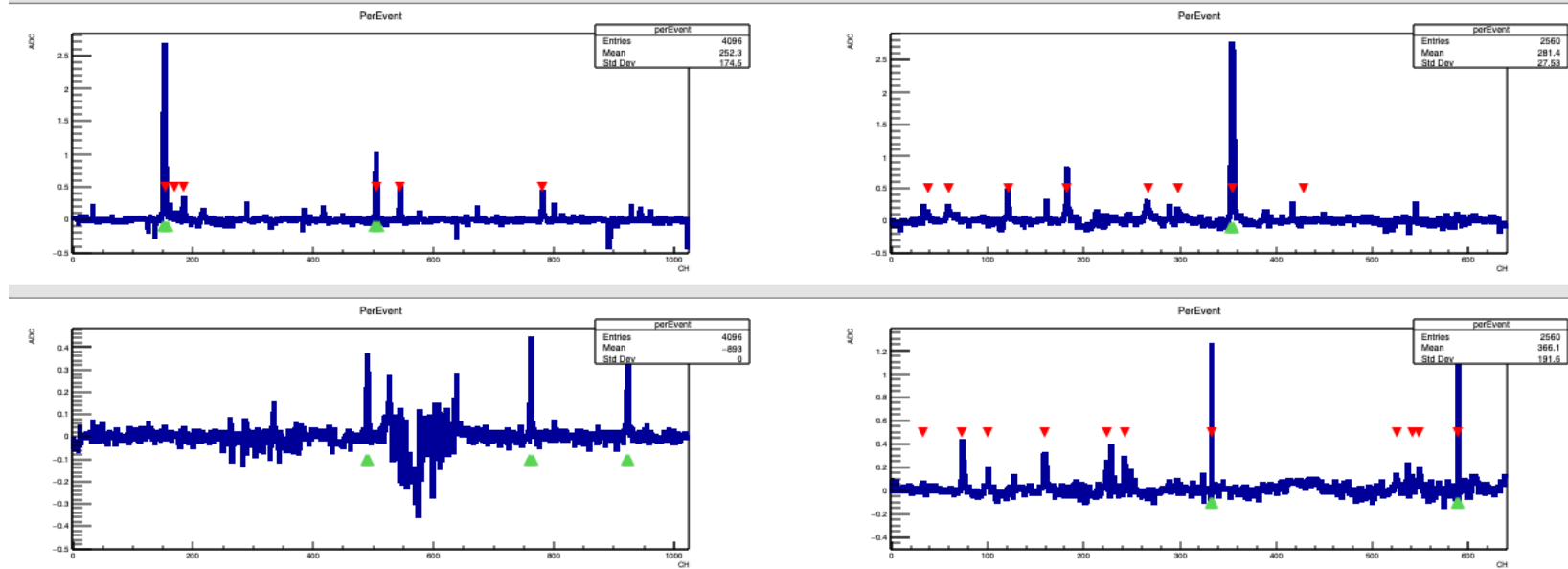


New Method

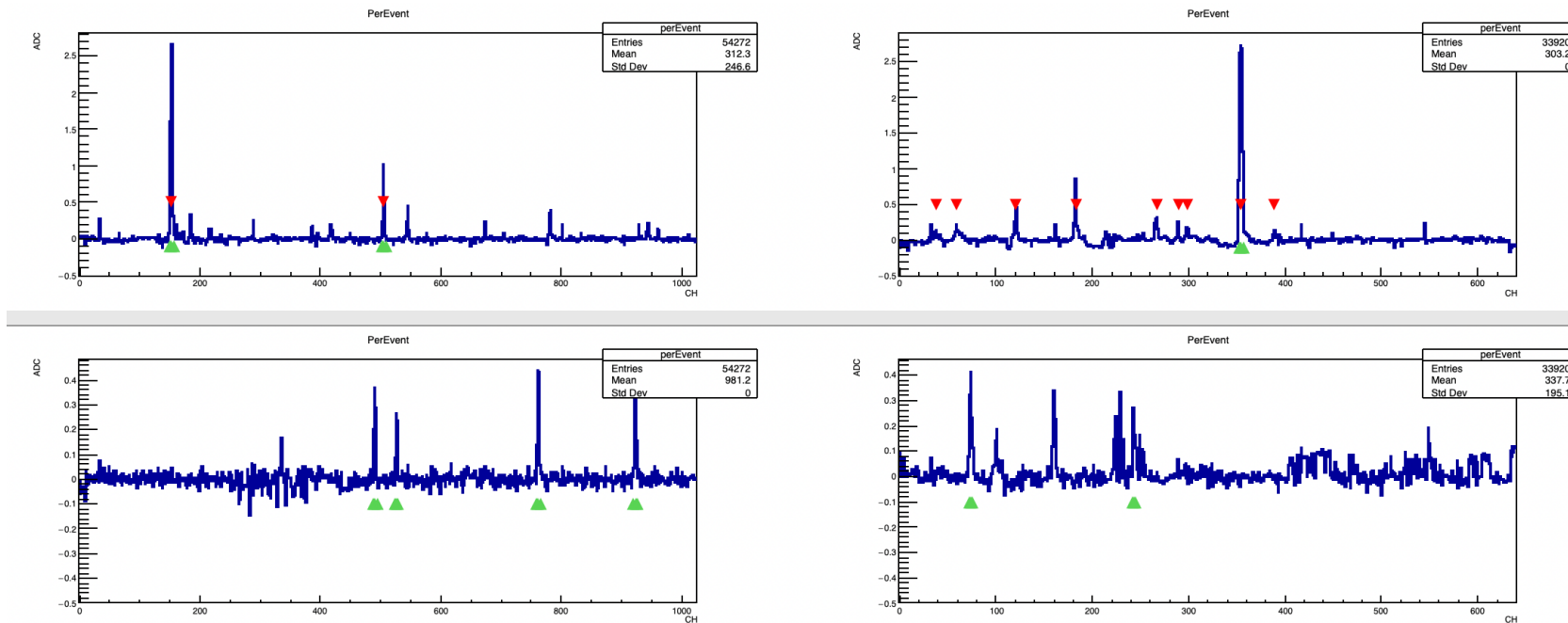


Run 4582, Event 721, Left Side

Old Method

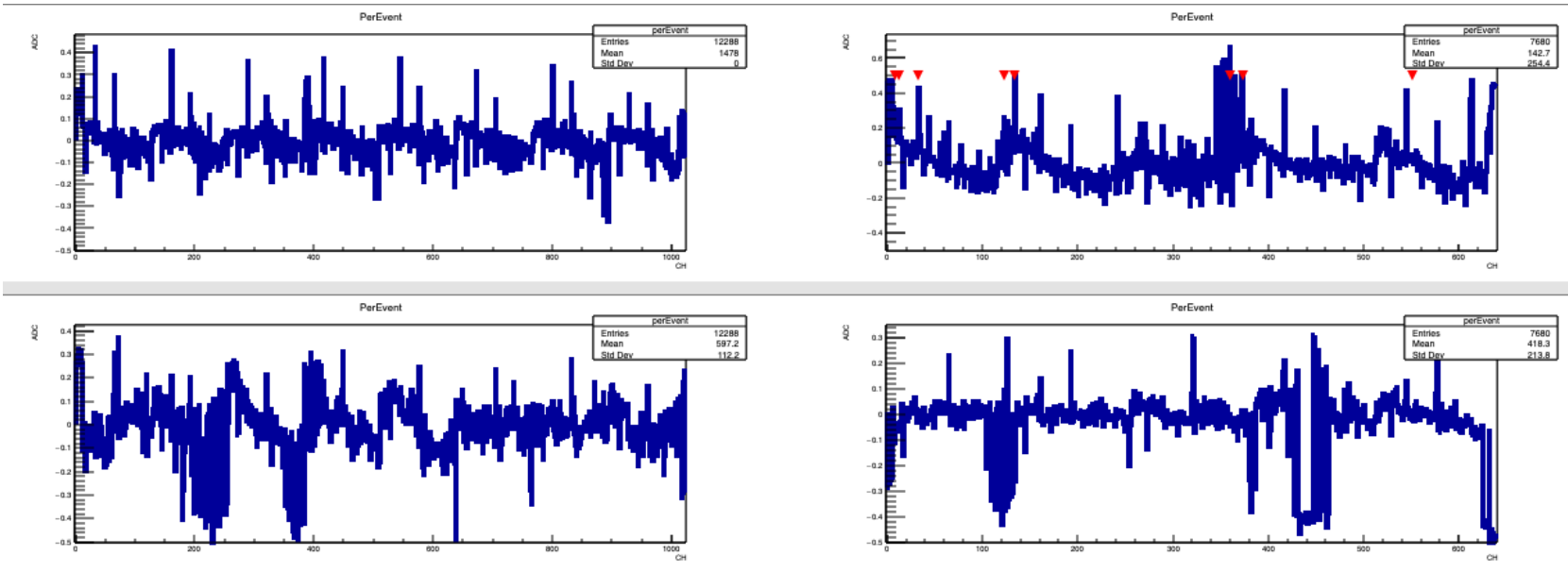


New Method

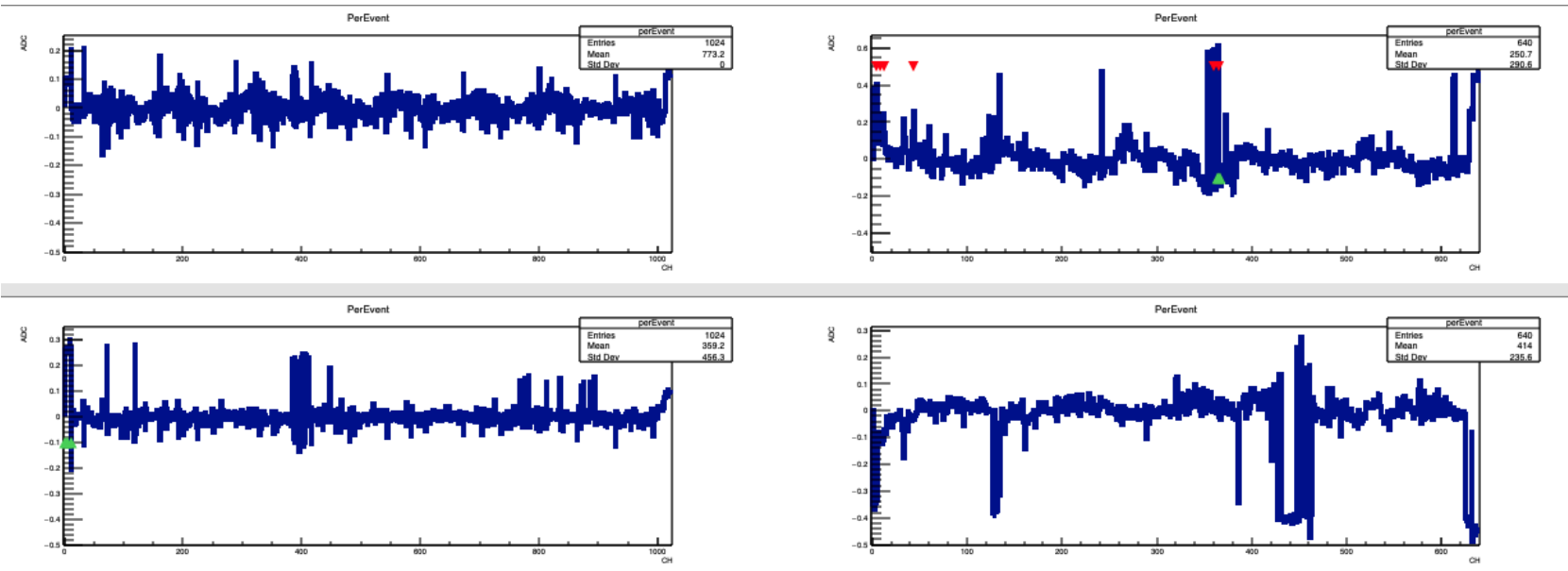


Run 4582, Event 721, Right Side

Old Method

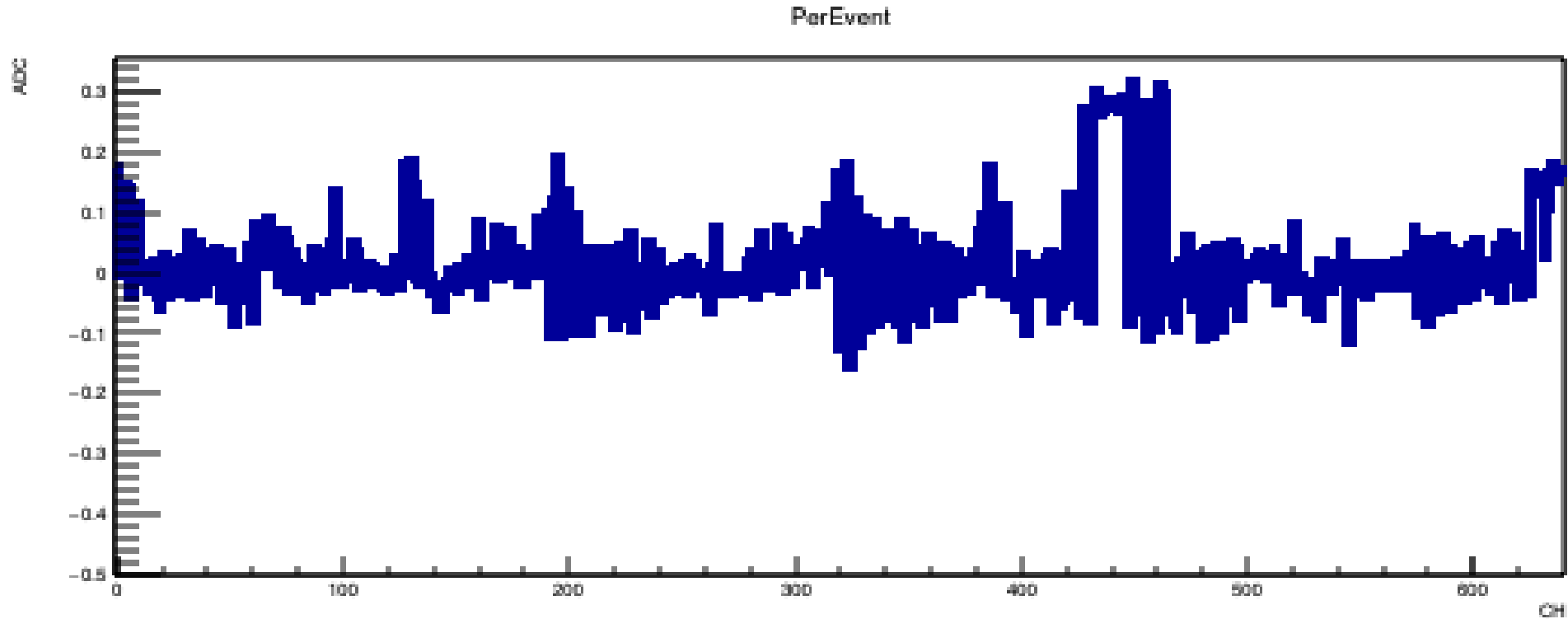


New Method



Needed Improvements

Run 4582, Event 8, GEM Right Top
Dispersive



Identifying square waves:

Thin ones are often falsely identified as hits.

The same square waves appear consistently throughout run 4582.

Run 4582, Event 907, Gem Right Bottom Dispersive

Filter out high noise events

Large waves traveling through data are unaffected by pedestals

Detectable with the same method used for pedestals

Not frequently detected as a hit

